

Comparison of a Multiphase Topology for Charging Energy Storage Elements

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Abstract – The paper presents comparison of multiphase converters for charging energy storage elements. Some of the main characteristics of these converters are compared. A simplified method for examination is developed. It is studied the option for powering the converters with current sources. The spectral characteristics of the current through the energy storage element for six-phase and seven-phase converter are compared.

Keywords – Charging Station, Multiphase Converter, Energy Storage, Resonant Converters.

I. INTRODUCTION

The design of powerful DC sources is based on three-phase systems in order for achieving higher output power and lowering the output voltage ripples as well. There are many researches which describe DC modules working in parallel. When DC-DC converters are connected in parallel, a major problem is equalizing the currents between the individual modules [1, 2, 3]. This problem is solved by connecting in parallel converters which are powered by current sources [4, 5]. Asymmetrical operation mode between the parallel connected modules influences the load current [6, 7]. Adjusting the load current is achievable by simultaneous adjusting all the output currents or by adjusting the current of one or several modules. During the process of adjustment the harmonic spectrum of the output current or voltage changes. The harmonic spectrum depends on the operation modes and the number of the modules. The proposed paper studies exactly this case. When using high-frequency modules [8], the spectral characteristics will be influenced by the parasitic elements in the circuit.

II. MULTIPHASE TOPOLOGIES AND MODELS

Fig. 1 shows a block diagram of a multi-phase converter working on a common load. The blocks Converter 1 to Converter N could be DC-DC converters (most often buck converters) or resonant converters [9]. Resonant converters are suitable as an intermediate part of a DC-DC converter. They are used when a galvanic isolation between the input and the output is needed. This is achieving with connecting the primary side of a high frequency transformer to the output of the resonant converter. To the secondary side of the transformer a high frequency rectifier is connected. In that case, the output of the rectifier is the output of the converter (Converter 1 to Converter N). The requirement for simultaneous work of rectifiers with connected in

parallel outputs is supplying them through current source. In that case, the resonant converters work exactly as current sources with current shape close to sine wave.

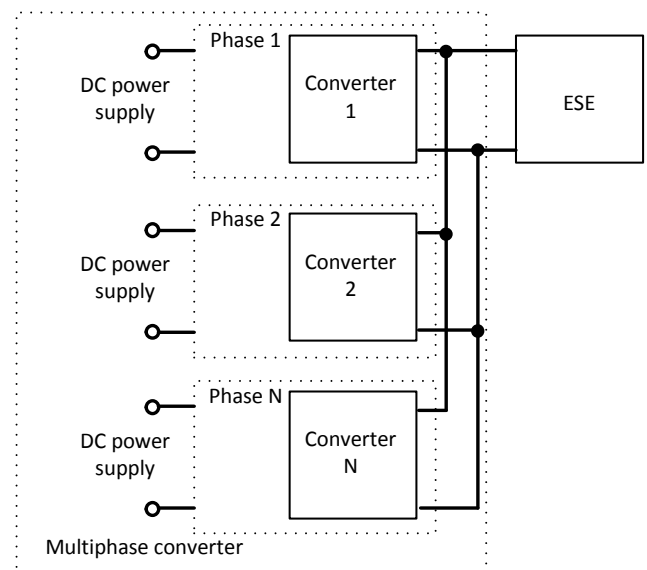


Fig. 1. A multiphase converter.

In this work, six-phase and seven-phase converters are studied and compared. For clarifying some of the features in the working principle of the six-phase converter, it could be described as two three-phase converters (converter A and Converter B) as it is shown in Fig. 2.

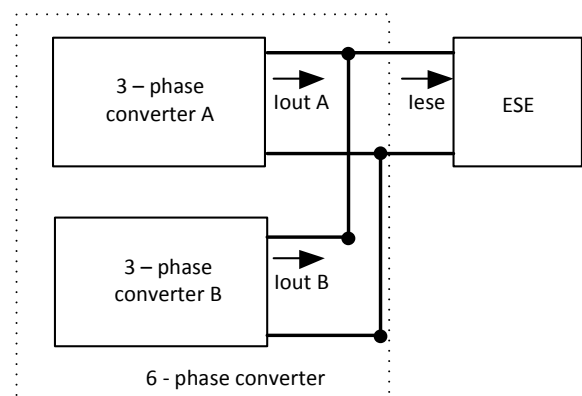


Fig. 2. Two three-phase converters operating as one six-phase.

The phase-shift of the currents for converter A is $2\pi/3$. For converter B, the currents phase-shift is the same. Therefore, the total phase-shift for the currents between the

converters is $2\pi/6$. In that way, the phase-shift between the serial phases is $2\pi/6$. A simulation model in LTSpice was developed.

In Fig. 3, the simulation model is shown. The sinusoidal current sources for converter A are numbered as I1, I3 and I5. Respectively, for converted B, they are numbered as I4, I6, and I2.

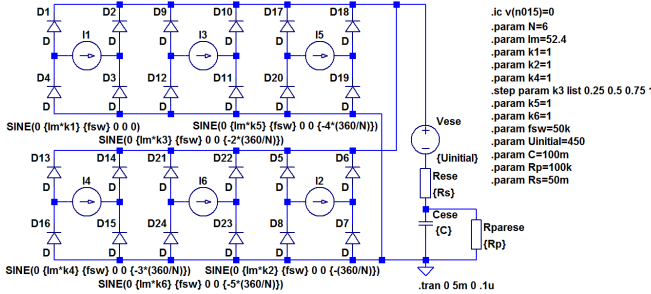


Fig. 3. LTSpice model for six-phase converter.

In the simulation model, the amplitudes of the currents of the individual sources change by the coefficients $k1 \div k6$.

Fig. 4 shows the following:

- $I(I1)$, $I(I3)$ and $I(I5)$ – three-phase currents for converter A and $I(I4)$, $I(I6)$ and $I(I2)$ – three-phase currents for converter B);
- output current of converter A ($I(D1)+I(D2)+I(D9)+I(D10)+I(D17)+I(D12)$);
- the output current from the converter B ($I(D5)+I(D6)+I(D13)+I(D14)+I(D21)+I(D22)$);
- the current through the energy storage element (Cese).

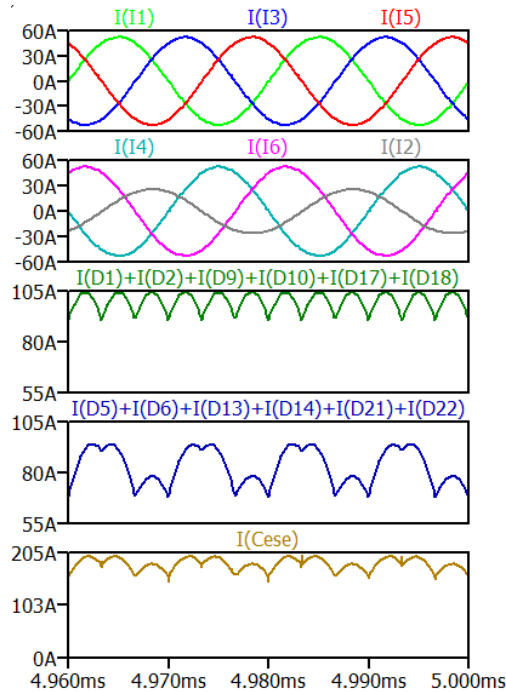


Fig. 4. Currents for three - phase converters A and B.

From the time diagrams shown in Fig. 4 it can be seen the currents asymmetry in converter B and how this currents

influences the shape of the current trough the energy storage element.

A simulation model for seven-phase converter has been created. It is designed as single seven-phase module in which the individual sources are phase-shifted at $2\pi/7$. Six-phase and seven-phase converters have equal output currents through the energy storage element – 200 amps. The energy storage element is represented with simplified equivalent circuit by using the elements Vese, Rese and Cese.

By changing the coefficients $k1 \div k6$ with values from 0 to 1 the amplitude of the currents changes. For one of the current sources the coefficient k has constant value of $k = 0.5$. For the other one the coefficient k changes with the following values $k = 0.25$, $k = 0.5$, $k = 0.75$ and $k = 1$. Different combinations of two sources have been studied, where the amplitudes of the currents change. The combinations are presented in Table 1.

Simulation model has been developed for seven-phase converter as well. The current amplitudes of two sources have been changed as well. The combinations are also presented in Table 1.

TABLE 1. COMBINATION FROM AMPLITUDE'S CHANGINGS OF THE SOURCES

	ph ase s	k1	k2	k3	k4	k5	k6	k7
case №								
1	6	0.5	var	1	1	1	1	-
	7	0.5	var	1	1	1	1	1
2	6	0.5	1	var	1	1	1	-
	7	0.5	1	var	1	1	1	1
3	6	0.5	1	1	var	1	1	-
	7	0.5	1	1	var	1	1	1
4	6	0.5	1	1	1	var	1	-
	7	0.5	1	1	1	var	1	1
5	6	0.5	1	1	1	1	var	-
	7	0.5	1	1	1	1	var	1

III. SIMULATION RESULTS

Results from simulation researches are shown below. Fig. 5 shows spectral characteristics of the current through the energy storage element when a six-phase converter is used. For better representation of the amplitudes of each of harmonics, the DC component with frequency of 0Hz is not shown. The results shown in Fig. 5 are related to case 1 from table 1.

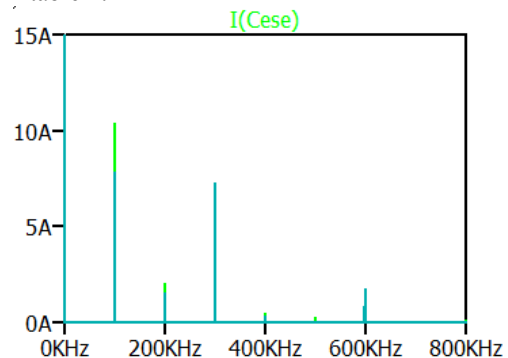


Fig. 5. Harmonic components in the output current for six - phase converter and case 1.

Fig. 6 illustrates how the amplitudes of the 2-nd, 4-th and the 6-th harmonic change when the current amplitude of one of the current sources changes for four values of the coefficient k as follows: $k=0.25$, $k=0.5$, $k=0.75$ and $k=1$.

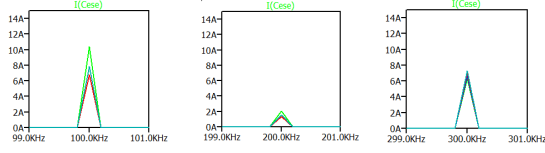


Fig. 6. Detailed visualization of harmonics from Fig. 5.

In Fig. 7 and Fig. 8 are shown results from the harmonic analysis of the current through the charged energy storage element in case 3.

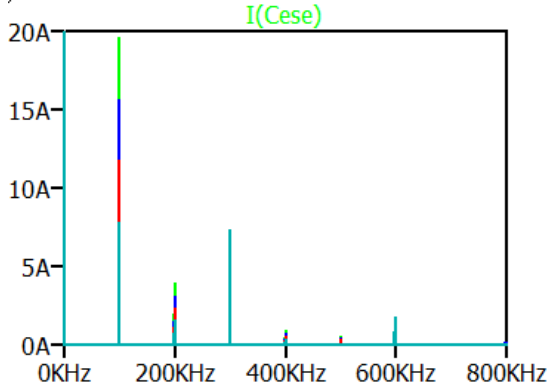


Fig. 7. Harmonic components in the output current for six-phase converter and case 3.

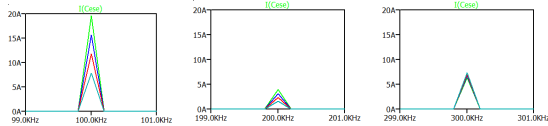


Fig. 8. Detailed visualization of harmonics from Fig. 7.

Fig. 8 shows the amplitudes of the 2-nd, 4-th and the 6-th harmonic from fig. 7. Comparing the results from Fig. 6 and Fig. 8 it can be seen that the amplitude of the 2-nd harmonic increases in case 3 (table 1). This amplitude is approximately twice as bigger in case 3 compared to case 1. Meanwhile the 6-th harmonic has approximately unchanged amplitude in both cases.

Similar researches for case 1 and case 3, related to seven-phase converter, has been done. Fig. 9 and fig.10 shows a spectral characteristic about case 1. In Fig. 11 and fig. 12, a spectral characteristic about case 3 is shown. Fig. 10 shows the amplitudes of the 2-nd, 4-th and the 6-th harmonic from fig. 7.

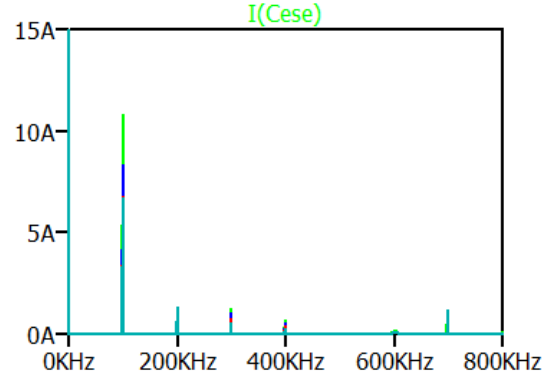


Fig. 9. Harmonic components in the output current for seven-phase converter and case 1.

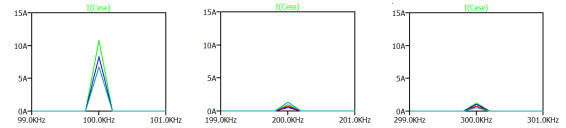


Fig. 10. Detailed visualization of harmonics from Fig. 9.

Comparing the results from Fig. 9 and Fig. 11 it can be seen that in case 3, the second harmonic has higher amplitude in contrast the same harmonic but for case 1.

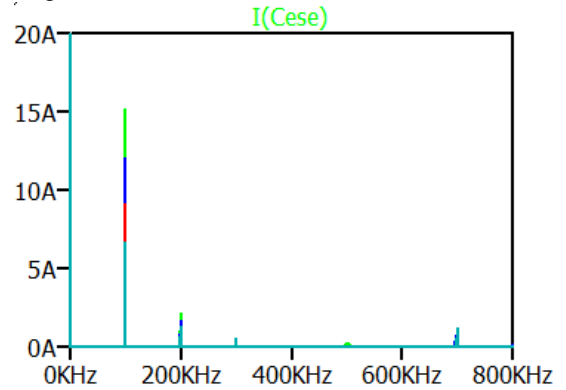


Fig. 11. Harmonic components in the output current for seven-phase converter and case 3.

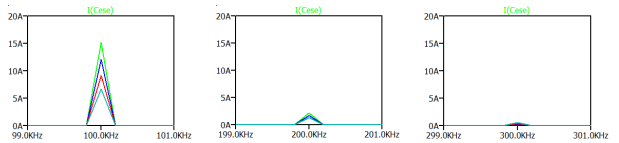


Fig. 12. Detailed visualization of harmonics from Fig. 11.

II. CONCLUSION

Investigation of multiphase converters, powered by current sources is an interesting problem in terms of their application in DC converters with higher output currents. The number of the modules influences the harmonic spectrum. The research has been done with purpose of determining the change in the amplitude of the individual harmonics in the output current when the

amplitude of the current of the supplying source changes. The obtained results can be used for proper selection of the filtering elements in the output of the rectifiers or for choosing what an algorithm has to be followed in order for changing the amplitudes of the AC supplying sources. In case of using the presented converters for charging energy storage elements, the need of filtering elements in the output of the converter can be estimated from the presented spectral characteristics. This is possible when the energy storage elements can operate with high frequency currents without this significantly to influence their exploitation characteristics.

Comparing the obtained results of examination of six-phase and seven-phase topologies the following conclusions can be made:

- The harmonic spectrum significantly depends on which phase the current amplitude has changed.
- When the current amplitudes of two contiguous phases change simultaneously (case 1), the second harmonic has lower amplitude compared with the case when the current amplitude of two inconsistent phases changes (case 3).
- The change of amplitudes of the 4-th and the 6-Th harmonic are approximately the same in both cases (case 1 and case 3).
- For the topology of the seven-phase converter it can be seen that the amplitudes of the 4-th and the 6-th harmonic do not change significantly and have lower values compared with the six-phase converter.

From the research, it can also be seen that with the seven-phase converter the specter of harmonics after the second harmonic have lower amplitudes in comparison with the six-phase converter. This makes it more suitable for using in multiphase systems. A drawback of the seven-phase converter is the more complex control system and that it has to be designed as a single converter (module). From the other hand, the six-phase converter could be designed as two three-phase converters with proper phase-shifting between the control systems for the two modules.

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